Hardness-Aware Contrastive Learning With Data Condensation for Open-Set Hyperspectral Image Classification

Hanyue Wang, Wei Liu, Jing Zhao, Xianghai Cao, *Member, IEEE*, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—In recent years, hyperspectral image classification (HSIC) has been widely applied in precision agriculture, environmental monitoring, and remote sensing. However, the frequent emergence of unknown categories, such as novel land-cover types and anomalous regions, poses significant challenges to traditional closed-set classifiers. Open-set classification seeks to accurately recognize known classes while rejecting unknown ones, but existing methods are often limited by high intraclass spectral variability, insufficient exploitation of hard samples, and imprecise class representations. To overcome these challenges, this article proposes an open-set detection framework called HACL-DC, which combines hardness-aware contrastive learning with soft-label-weighted data condensation. The framework consists of three key components. First, a similarity contrastive weighting module enhances contrastive learning by assigning higher weights to hard intraclass positives and interclass negatives through linear transformations combined with sigmoid and inverse-sigmoid functions, promoting compact intraclass features and stronger interclass separation. Second, a soft-label-weighted condensation algorithm (SWCA) constructs compact etalon sets for each known class by leveraging soft-label assignment, dual-parameter gradient optimization, and adaptive etalon resetting to capture fine-grained class distributions and local density information. Finally, weighted similarities between test samples and class etalons are computed using etalon influence scores, enabling accurate known-class classification and effective detection of out-of-distribution (OOD) samples. Extensive experiments on three public hyperspectral datasets demonstrate the effectiveness and robustness of the proposed HACL-DC framework.**

***Index Terms*—Contrastive learning, data condensation, hyperspectral images, open-set classification.**

I. Introduction

HYPERSPECTRAL imaging is an advanced sensing technology that integrates imaging and spectral analysis. By capturing continuous narrowband spectral signatures along with spatial details, it provides tens to hundreds of fine spectral channels per pixel, enabling precise extraction of material characteristics for both qualitative analysis and quantitative retrieval [1], [2]. Owing to its rich spectral–spatial information, hyperspectral imaging has become widely used in remote sensing, precision agriculture, environmental monitoring, medical diagnosis, food safety, and mineral exploration [3], [4]. Among these applications, hyperspectral image classification (HSIC) plays a fundamental role. However, challenges such as high dimensionality and strong correlation among spectral bands often prevent traditional image classification methods from achieving satisfactory performance [5].

Early studies primarily relied on traditional machine learning [6]. Handcrafted feature engineering combined with classical classifiers—such as support vector machines [7], k-nearest neighbors [8], and maximum likelihood classifiers [9]—dominated the field [6]. Dimensionality-reduction techniques like PCA [10] and LDA [11] were commonly used to mitigate the curse of dimensionality. Nevertheless, these methods depend on manual feature extraction and struggle to model complex nonlinear spectral–spatial relationships in hyperspectral data, resulting in suboptimal performance under noise, spectral mixing, or complex scenes [12], [13].

With the rise of deep learning, HSIC has achieved substantial progress. CNN-based models automatically learn hierarchical spectral–spatial representations and outperform traditional approaches [14], [15]. Advanced architectures such as 3-D CNNs and CNN–RNN hybrids further enhance spectral–spatial dependency modeling [16], [17]. Transformer-based models and attention mechanisms enhance the ability to model long-range dependencies and improve feature discriminability [18]. Other paradigms—such as graph neural networks (GNNs) [19] and generative adversarial networks (GANs) [20]—enhance robustness to spatial topology and limited labeled data. Together, these approaches demonstrate the effectiveness of deep learning in capturing complex spectral–spatial structures from hyperspectral data.

Despite these advances, most hyperspectral classification models implicitly assume that training and test samples come from the same distribution. In practice, hyperspectral scenes often contain unseen land-cover types or significant spectral variations not present in the training set [21]. These out-of-distribution (OOD) samples deviate from the training distribution [22], [23], causing existing classifiers to misclassify them as known classes (see Fig. 1), which may lead to serious errors in environmental monitoring or resource assessment. Therefore, reliable OOD detection is essential for practical hyperspectral applications.

Received 19 May 2026; revised 16 August 2026; accepted 17 September 2026. Date of publication 22 September 2026; date of current version 5 October 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62176199, in part by the Key Research and Development Projects of Shaanxi Province under Grant 2024NC-YBXM-212. *(Corresponding author: Xianghai Cao.)*

Hanyue Wang, Wei Liu, Xianghai Cao, and Licheng Jiao are with the School of Artificial Intelligence, Xidian University, Xi'an 710126, China (e-mail: xianghaicao@hotmail.com).

Jing Zhao is with the School of Artificial Intelligence, Xidian University, Xi'an 710126, China, and also with China United Network Communications Group Company Ltd., Shaanxi Branch, Xi'an 710075, China.

Digital Object Identifier 10.1109/TGRS.2026.3736209

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

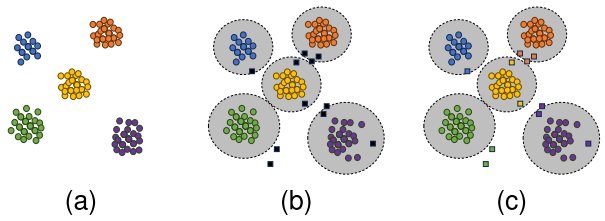


Fig. 1. Illustration of OOD classification. (a) Training data containing only ID samples. (b) Test data containing unknown OOD samples. (c) Incorrect classification of OOD samples.

To address this problem, open-set remote sensing image classification has emerged as an effective paradigm. Its objective is to train a model that can correctly classify samples from known categories while reliably rejecting samples from unknown ones. Existing approaches can be broadly grouped into two groups: postprocessing approaches [24], [25] and training-based approaches. Among the latter, reconstruction-based methods [26], [27], [28], [29], [30], [31] and distance-based methods [22], [32], [33], [34], [35], [36] are the most widely studied approaches.

Postprocessing-based methods operate on pretrained closed-set classifiers. They implement open-set rejection by adjusting model outputs through confidence-based thresholds. For example, Liu et al. [24] employed box-plot statistics to derive classwise rejection thresholds, and Alwadei et al. [25] combined ConvMixer with an energy-based model to enhance feature discrimination. These methods are easy to deploy, but their performance is highly sensitive to threshold selection and often unstable in complex scenarios.

Reconstruction-based methods leverage the representation learning and reconstruction capabilities of generative models, based on the key assumption that the model can accurately reconstruct samples from known classes seen during training, whereas samples from unknown classes, which are not learned by the model, yield significantly higher reconstruction errors. Representative works include MDL4OW proposed by Liu et al. [26], geometric–spectral reconstruction methods that jointly exploit hyperspectral and LiDAR data [27], and spectral–spatial reconstruction frameworks that operate on learned latent representations [28]. Ji et al. [29] incorporated GAN-based uncertainty estimation, and Ma et al. [30] utilized source-domain reconstruction for target-domain unknown detection. Xi et al. [31] proposed HyperCASR, a spectral–spatial open-set recognition framework with class-aware semantic reconstruction using class-specific autoencoders. However, these approaches depend heavily on the generative capacity of the model and degrade when known classes exhibit strong intraclass spectral variability.

Distance-based methods have become increasingly popular because of their clear interpretation and compatibility with hyperspectral data. They model feature distributions by learning feature centers, class boundaries, or distance metrics such as cosine or Mahalanobis distance. During inference, samples far from known-class distributions are flagged as unknown. For example, Xie et al. [32] used a CNN with contrastive clustering to learn discriminative prototypes. Xu et al. [33] improved feature separability through a class-anchoring strategy. Chen et al. [34] introduced the fractional Fourier transform to enhance class differences, while Feng et al. [35] used hyperspherical prototype learning to make class features more separable. Qin et al. [36] incorporated semantic information through a language-enhanced contrastive framework.

Although promising, existing distance-based methods still face three major limitations. First, hyperspectral data often exhibit significant intraclass variability caused by illumination changes, atmospheric interference, and mixed pixels, resulting in sparse and overlapping feature distributions. Second, existing contrastive-learning-based methods usually treat all sample pairs equally, insufficiently emphasizing hard positive and hard negative pairs, which limits feature compactness and interclass separability. Third, representing each class using a single prototype or a KNN-averaged representation cannot effectively capture complex multimodal structures and is sensitive to outliers, especially under limited training samples.

To mitigate these limitations of existing methods, we introduce contrastive learning to improve feature compactness and class separability. Conventional contrastive learning, however, mainly optimizes cosine similarity and does not account for whether hard sample pairs are sufficiently optimized [37]. To address this issue, we propose a hardness-aware similarity contrastive weighting mechanism (HSCWM) that assigns higher weights to hard positive and negative pairs.

Furthermore, instead of representing each class using a single prototype, we propose a soft-label-weighted condensation algorithm (SWCA) to construct representative etalons that better capture local distribution structures. SWCA adaptively generates a set of representative etalons for each class based on local feature density and similarity relationships, assigning larger weights to informative samples. This produces more reliable and structurally aware class representations.

HSCWM focuses on feature optimization by emphasizing difficult sample relationships during contrastive learning, thereby producing more compact intraclass distributions and clearer interclass margins in the hyperspherical space. Based on these optimized features, SWCA further performs structure-aware condensation to construct representative etalons that capture local distribution characteristics and multimodal intraclass structures. Therefore, HSCWM improves the discriminability of feature representations, while SWCA enhances the reliability of class representation and distance measurement. Their combination jointly strengthens the robustness of OOD detection in complex hyperspectral scenarios.

Based on these designs, we develop HACL-DC, an OOD detection framework that integrates HSCWM and SWCA. Experiments conducted on the Indian Pines, WHU-Hi-LongKou, and WHU-Hi-HanChuan benchmarks demonstrate that HACL-DC consistently improves OOD detection performance and outperforms representative existing methods.

The major contributions of this work are summarized as follows.

1) To address insufficient optimization of difficult sample relationships in existing contrastive-learning-based OOD methods, a HSCWM is proposed to adaptively

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

emphasize hard positive and negative pairs, improving intraclass compactness and interclass separability.

2) To overcome the limitations of single-center class representation, a SWCA is developed to generate structurally representative etalons through density-aware weighting and soft-label assignment, enabling more accurate modeling of complex intraclass hyperspectral distributions.
3) By projecting features onto a hyperspherical space and jointly applying HSCWM for feature optimization and SWCA for etalon refinement, HACL-DC establishes an integrated pipeline encompassing feature optimization, etalon condensation, and weighted distance measurement for robust OOD detection.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the proposed HACL-DC framework in detail. Section IV reports the experimental results on three hyperspectral image datasets. Finally, Section V concludes this article.

II. Related Work

A. OOD Detection

In multiclass recognition tasks, the training data are typically assumed to follow an in-distribution (ID) source $P_{in} = \{(x_i, y_i\}_{i=1}^{N}$, where each sample–label pair (x_i, y_i) is associated with one of the C known categories, i.e., $y_i \in \{1, 2, \ldots, C\}$. During inference, the test stream may contain samples drawn from an OOD source P_{out}, whose statistical characteristics differ substantially from those of P_{in}. This distributional discrepancy motivates the need for OOD detection, which is intended to identify samples that do not belong to any known class [26].

OOD detection is commonly formulated as a binary decision task driven by a detection score assigned to each test sample. A sample is classified as an ID instance when its score exceeds a predefined threshold; otherwise, it is designated as an OOD sample. For fair comparison across different methods, this threshold is typically calibrated using ID training data. A standard practice is to set the threshold such that the true positive rate (TPR) on the ID set reaches 95%.

B. Contrastive Learning

Contrastive learning is a powerful self-supervised representation learning paradigm that aims to construct an embedding space in which similar samples (positive pairs) are pulled close together, while dissimilar samples (negative pairs) are pushed apart [37]. By explicitly comparing sample relationships, it enables the model to capture the intrinsic structure of the data. In this work, contrastive learning is employed to maximize the cosine similarity among samples from the same class and minimize the similarity across different classes, thereby yielding highly discriminative feature representations.

C. Hyperspherical Embeddings

Hyperspherical embeddings refer to feature representations constrained to lie on the surface of a hypersphere with radius r_h. A common approach to obtaining such embeddings is to apply L_2-normalization, which scales each feature vector to unit norm, typically with $r_h = 1$, thereby projecting it onto the hyperspherical manifold. In our framework, features are first normalized in this manner and then processed within the hyperspherical space [38]. On top of this normalized representation, we integrate the proposed similarity-weighted contrastive mechanism into the contrastive learning stage. This integration enhances the discriminative power of the learned features by encouraging more structured separation on the hypersphere.

III. Methodology

The overall framework of the proposed HACL-DC method is shown in Fig. 2. During training, hyperspectral image patches are first processed by the feature extraction network to obtain feature maps, which are subsequently projected onto a hyperspherical manifold through a projection head. The HSCWM module is then employed to perform similarity-weighted contrastive learning, emphasizing hard positive and hard negative pairs to simultaneously reduce intraclass variability and enlarge interclass separability. Thereafter, the SWCA module initializes up to K representative etalons, whose positions and influence weights are iteratively refined via gradient-based optimization. Using these refined etalons, the model is further fine-tuned to establish stable and discriminative class representations.

During inference, both the etalon positions and their influence weights learned from the training set are used. For each test sample, its minimum weighted distance to the etalon set is computed and converted into a similarity score. Distributional classification is subsequently performed by comparing this score with a predefined threshold, enabling the identification of both ID and OOD samples.

The overall training procedure of the proposed HACL-DC framework is summarized in Algorithm 1. The detailed designs of each module are presented in Sections III A–III C.

A. Hardness-Aware Similarity-Weighted Contrastive Optimization

Let (x_i, x_j) denote an intraclass pair and $(x_i, x_{j'})$ an interclass pair. Let c_w and c_b denote their corresponding cosine similarities. Let f represent the feature embedding after hyperspherical projection. The cosine similarities are computed as $c_w^i = \hat{f}_i \cdot \hat{f}_j$ and $c_b^i = \hat{f}_i \cdot \hat{f}_{j'}$, where $\hat{f}$ is the L_2-normalized feature vector.

Assuming equal contributions from intraclass and interclass similarities, the contrastive loss is expressed as

$$\mathcal{L} = log\left(\sum_{i=1}^{n} exp\left(c_b^i/\tau\right)\right) - log\left(\sum_{j=1}^{m} exp\left(c_w^j/\tau\right)\right) \quad (1)$$

where n and m denote the numbers of interclass and intraclass pairs, respectively, and τ is a temperature parameter.

Ideally, samples from the same class should cluster tightly around their centroid, while different classes should exhibit large interclass margins. However, in practice, the distribution often resembles the pattern illustrated in Fig. 3(a): a portion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

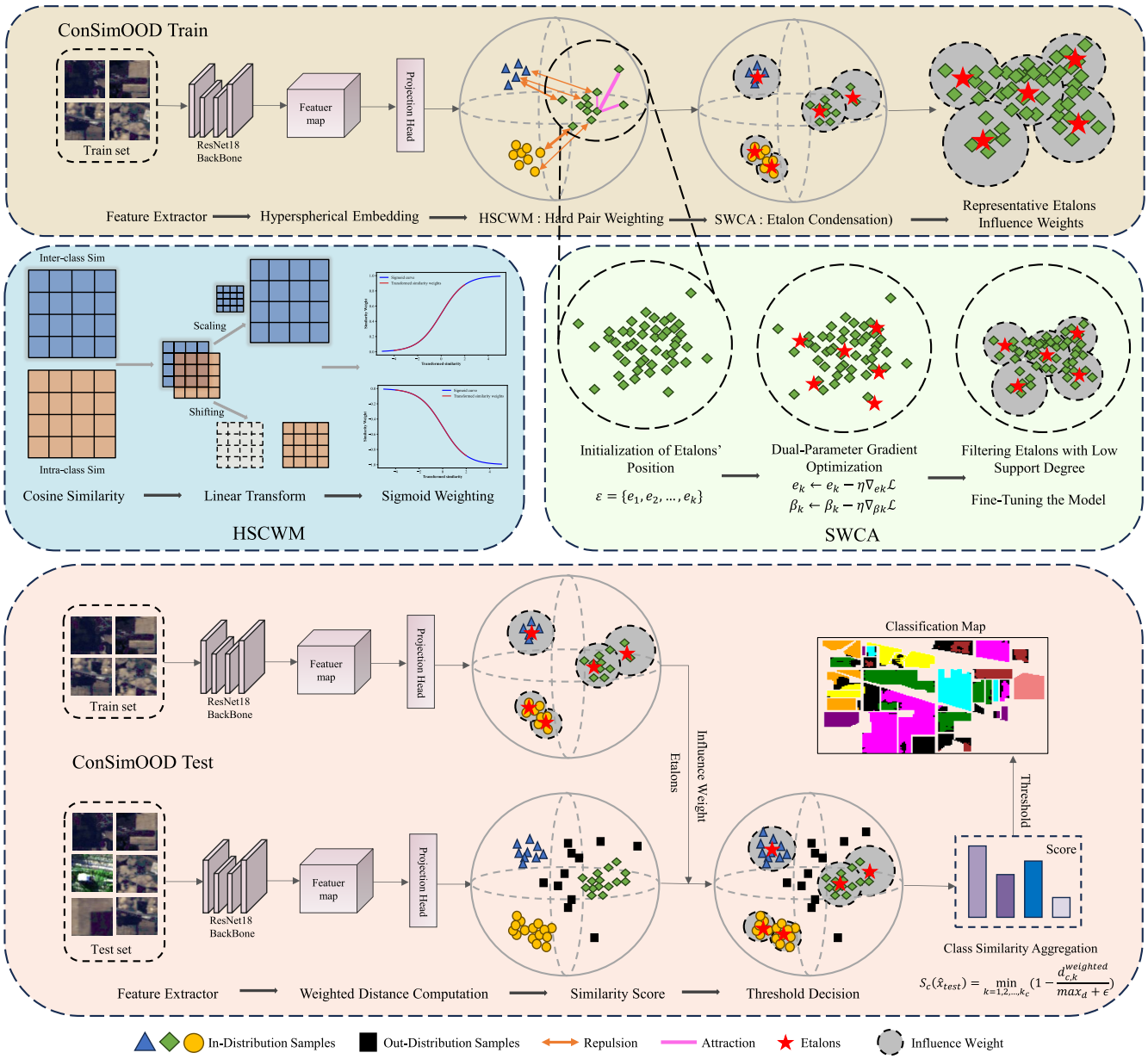


Fig. 2. Overall architecture of the proposed HACL-DC framework. During training, hyperspectral patches are processed by the feature extractor and projection head, followed by the HSCWM module for hardness-aware similarity weighting and the SWCA module for representative etalon condensation. The learned etalons and influence weights are then transferred to the inference stage, where weighted similarity computation and threshold-based decisions are performed for open-set classification.

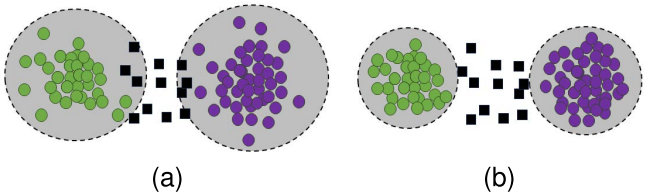


Fig. 3. (a) Contrastive learning without optimization and (b) contrastive learning with optimization. Circles denote ID samples, and squares denote OOD samples. After optimization, the ID features form a more compact cluster, enabling clearer separation from OOD samples.

of samples forms the expected compact cluster near the centroid, whereas others remain loosely scattered outside the core region. These outlying samples substantially affect both the minimum intraclass distance and the maximum interclass distance.

However, the conventional contrastive objective in (1) assigns identical weights to all positive and negative pairs, implicitly assuming that they provide equivalent optimization information. From the metric learning perspective, easy pairs already approach the desired similarity distribution and contribute limited gradients during optimization, whereas hard positive pairs with low similarity and hard negative pairs with high similarity usually lie near class boundaries, exhibiting larger deviations and providing more informative optimization signals. Therefore, uniform weighting may cause the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Overall Procedure of HACL-DC

Input: Training set $\mathcal{D}$, number of etalons K, maximum epochs E, maximum iterations T

Output: Optimized feature representation, refined etalon set $\mathcal{E}^*$, learned influence factors β^*

```
Initialize model parameters
for epoch = 1 to E do
    Sample a mini-batch B
    Extract feature embeddings
    Normalize feature embeddings onto the hypersphere
    Compute positive and negative cosine similarities
    Compute hardness-aware similarity weights according to Eqs. (2)–(3)
    Compute weighted contrastive loss according to Eq. (4)
    Update model parameters
end for
Freeze the feature extractor
Extract normalized features for all training samples
for each known class do
    Randomly initialize K etalons
    Initialize β_k = 1 and p_k = 0
    for t = 1 to T do
        Compute distances between features and etalons
        Compute soft assignment weights according to Eq. (5)
        Compute ELBO objective according to Eq. (6)
        Update etalon locations according to Eq. (11)
        Update influence factors according to Eq. (13)
        Anneal temperature according to Eq. (14)
        Compute etalon supports
        if t > 0.1T and mod(t, 10) = 0 then
            if p_k < θ then
                Reset inactive etalons
            end if
        end if
    end for
    Remove redundant etalons according to Eq. (15)
end for
return optimized feature representation, E*, and β*
```

model to be dominated by redundant easy samples rather than focusing on discriminative hard pairs. By adaptively increasing the contribution of these hard samples, HSCWM enables the model to focus on ambiguous feature regions and achieve improved intraclass compactness and interclass separability.

To mitigate this issue, we introduce HSCWM, which reweights cosine similarities according to sample difficulty, assigning larger weights to hard positive and hard negative pairs. In the embedding space, sample difficulty is naturally reflected by feature similarity. Specifically, for positive pairs from the same class, those with relatively low cosine similarity are regarded as hard positive pairs because their features are not sufficiently compact. In contrast, for negative pairs from different classes, those with relatively high cosine similarity are regarded as hard negative pairs since they are more likely

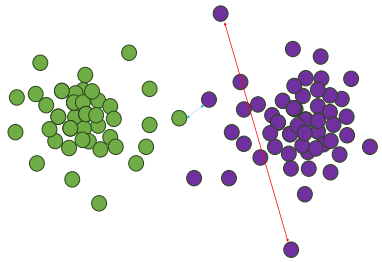


Fig. 4. Illustration of hard positive and hard negative sample pairs in the embedding space. Red double-headed arrows denote hard positive pairs with relatively low intraclass similarity, while blue double-headed arrows denote hard negative pairs with relatively high interclass similarity.

to be confused during classification. These hard pairs typically lie near class boundaries and therefore contribute more to learning discriminative feature representations. As illustrated in Fig. 4, hard positive pairs are connected by red double-headed arrows, while hard negative pairs are connected by blue double-headed arrows. Based on these observations, the weighting is formulated as a function of cosine similarity. Prior to weighting, the similarity values are linearly transformed using a slope parameter S and an intercept parameter I, aligning the transformed range with the sensitive region of the subsequent sigmoid-based weighting function.

The original cosine similarity lies within $[-1, 1]$. The slope S controls the scaling of this range, thereby influencing how rapidly the weighting factor grows with respect to similarity. Before the similarity values are passed to the weighting function, they are linearly adjusted via multiplication with S, effectively mapping the range from $[-1, 1]$ to $[-S, S]$. For instance, when $S > 1$, differences between hard and easy samples become more pronounced, allowing the weighting function to respond more sensitively to sample difficulty. The slope parameter is tuned so that most transformed similarities fall within the sigmoid's sensitive region (approximately $[-5, 5]$).

The intercept I shifts the scaled similarity distribution, determining which portion of the similarity range is assigned higher attention by the sigmoid function. Given the linear transformation $T = C \cdot S + I$, where C is the original cosine similarity, S the slope, and T the transformed similarity, the intercept I directly shapes the region in which the weighting function yields large responses. This enables HSCWM to precisely target and strengthen hard samples, thereby improving optimization effectiveness for both intraclass compactness and interclass separability.

Given the intrinsic difference between the optimization objectives of within-class and between-class similarity (the former should be maximized, while the latter should be minimized), the linear transformation parameters for the two types of similarity must be optimized independently. We define a linear transformation for within-class similarity as $T_W = C_w \cdot S_w + I_w$, and for between-class similarity as $T_b = C_b \cdot S_b + I_b$. This ensures that the HSCWM mechanism can accurately accommodate the optimization needs of different sample types, jointly enhancing within-class compactness and between-class separability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

After linear transformation, we employ sigmoid-based weighting due to its bounded and nonlinear characteristics. Specifically, sigmoid maps the transformed similarity values into $(0, 1)$, providing stable weight allocation while avoiding excessive emphasis on extremely difficult samples. Its nonlinear response allows the model to distinguish samples with different difficulty levels: hard samples located in the sensitive region receive larger weight variations, whereas easy samples obtain relatively saturated weights. Moreover, the parameters S and I provide flexibility to align the activation region with different similarity distributions. Compared with linear weighting, sigmoid-based weighting offers a smoother and more robust strategy for emphasizing informative samples.

Considering that high-similarity interclass pairs are more likely to cause category confusion, we apply the sigmoid function to assign larger weights to hard negative samples

$$\phi(T_b) = 1 \Big/ \left(1 + e^{-T_b}\right) = 1 \Big/ \left(1 + e^{-C_b S_b - I_b}\right). \tag{2}$$

For within-class similarity, we use an inverted sigmoid function to assign higher weights to hard positive samples with low similarity, improving within-class compactness

$$\phi(T_w) = 1 \Big/ \left(1 + e^{T_w}\right) = 1 \Big/ \left(1 + e^{C_w S_w + I_w}\right). \tag{3}$$

According to the above weighting strategy, hard negative samples receive larger effective temperatures in the denominator of the contrastive objective, increasing their contribution to interclass separation. Meanwhile, hard positive samples are assigned smaller effective temperatures, strengthening the optimization pressure for intraclass aggregation. Therefore, HSCWM does not simply increase or decrease similarity values but adaptively modifies the optimization intensity according to sample difficulty. Both functions adapt the input range through linear transformation, ensuring precise and stable weight allocation. Ultimately, they help achieve a feature space characterized by tight within-class clusters and well-separated between-class distributions, providing a more robust foundation for OOD detection.

Accordingly, the contrastive loss in (1) is updated as

$$\mathcal{L} = \log\left(\sum_{i=1}^{n} \exp\left(\frac{C_b^i}{\tau + \tau e^{-C_b^i S_b - I_b}}\right)\right) - \log\left(\sum_{j=1}^{m} \exp\left(\frac{C_w^j}{\tau + \tau e^{C_w^j S_w + I_w}}\right)\right). \tag{4}$$

B. Soft-Label-Weighted Condensation for Class Representation

To overcome the limitations of conventional modeling strategies—such as relying on a single class center or average distance, which often fail to characterize the fine-grained and multimodal structure of hyperspectral data—we introduce SWCA. The proposed algorithm integrates a dynamic soft-assignment mechanism, a dual-parameter gradient optimization scheme, and an adaptive etalon-resetting strategy to construct a compact yet highly representative set of etalons for each known class. Here, an etalon refers to a condensed feature exemplar generated from a subset of semantically similar training samples. Rather than representing an entire class with a single prototype or class center, multiple etalons are maintained for each class to characterize diverse local semantic patterns and preserve intraclass structural variations.

Unlike conventional hard-assignment approaches that associate each sample with only one prototype, SWCA models the relationship between samples and etalons in a probabilistic manner. Specifically, each sample maintains soft associations with multiple etalons according to their feature similarities, enabling the model to capture complex intraclass distributions with multiple local modes and significant spectral variability. The resulting soft-label weights reflect the contribution of different local structures to the representation of each sample, thereby providing a more flexible and fine-grained characterization of hyperspectral feature distributions.

Meanwhile, each etalon is associated with an influence factor, which characterizes its effective coverage range and local density sensitivity in the feature space. Etalons located in dense and representative regions tend to learn smaller influence factors, thereby imposing stronger constraints on nearby samples. In contrast, etalons situated in sparse or boundary regions are assigned relatively larger influence factors to prevent unstable local structures from dominating the representation process. Through the joint optimization of soft-label assignments, etalon locations, and influence factors, SWCA progressively refines structurally representative etalons that simultaneously preserve compact core regions and adapt to complex boundary structures. As a result, the learned representations provide more reliable distance references for distinguishing ID samples from OOD samples in the subsequent open-set classification stage.

1) Algorithm Initialization: The initialization stage of SWCA defines the core parameters and the initial states of learnable variables to ensure both stable convergence and effective class representation. The input parameters include are as follows.

1) The preset number of etalons K, selected adaptively according to class size and data complexity.
2) The initial temperature parameter τ', which controls the fuzziness of soft assignment and is set to 1.0 by default.
3) The learning rate η to update the etalon positions and influence coefficients.
4) The temperature decay rate α, which determines the exponential decay of τ'.
5) The support threshold θ, used to identify invalid etalons, defined as 30% of the average support across all etalons.
6) The maximum number of iterations.

Each etalon initialization is performed by generating K random D-dimensional feature vectors $\mathcal{E} = \{e_1, e_2, \ldots, e_k\}, k = 1, 2, \ldots, K$, sampled from a standard normal distribution to ensure uniform dispersion. Each etalon is assigned an initial influence coefficient $\beta_k = 1.0$, which characterizes its local impact radius and density sensitivity.

All etalon support values are initialized as $p_k = 0, (k = 1, 2, \ldots, K)$, where the support is defined as the accumulated soft-assignment weights from samples to the etalon. This support measure serves to evaluate etalon validity and facilitates subsequent resetting of ineffective etalons.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

2) Soft-Label Assignment Mechanism: In each iteration, SWCA employs a soft-label assignment mechanism to compute the affiliation weight of each sample with respect to each etalon, enabling dynamic and continuous sample–etalon association. Given an intraclass feature set $X = \{x_1, x_2, \ldots, x_N\}, x_i \in \mathbb{R}^D$, where N is the number of samples in the class, the L_1 distance between each sample x_i and each etalon e_k is computed. The L_1 metric is adopted to mitigate the influence of outliers commonly present in hyperspectral data, thereby improving the robustness of distance-based modeling. This yields a distance matrix $\mathcal{D} \in \mathbb{R}^{N\times K}$, with elements defined as $d_{ik} = \|x_i - e_k\|_1$.

A temperature parameter τ' is then introduced to scale these distances $\tilde{d}_{ik} = -d_{ik}/\tau'$. A larger τ' produces a more uniform weight distribution—resulting in fuzzier soft labels—whereas a smaller τ' sharpens the distribution, making the assignments closer to hard labels. Applying a softmax function to the scaled distances yields the soft-assignment weight of x_i to etalon e_k, with $\epsilon = 10^{-8}$ included for numerical stability

$$w_{ik} = \frac{\exp\left(\tilde{d}_{ik}\right)}{\sum_{j=1}^{K}\exp\left(\tilde{d}_{ij}\right) + \epsilon}. \tag{5}$$

These weights satisfy $w_{ik} \in [0, 1]$, $\sum_{k=1}^{K} w_{ik} = 1$. The value w_{ik} represents the confidence with which sample x_i is associated with etalon e_k, forming the sample's soft-label representation. This mechanism transitions the modeling paradigm from hard assignment to soft, continuous association, enabling richer and more flexible characterization of intraclass feature structure.

3) Bi-Parameter Gradient Optimization: Based on the soft-assignment weights obtained in the E-step, SWCA updates both etalon locations e_k and their influence factors β_k via gradient descent. The optimization objective is to maximize the variational lower bound (ELBO) of the spherical Laplacian mixture model. Accordingly, the loss function is defined as the negative ELBO

$$\mathcal{L}' = -\text{ELBO} = -\left(\text{L}_1 + \text{L}_2 + \text{L}_3\right). \tag{6}$$

Prior term (L_1) assumes a uniform prior over etalons, i.e., $\pi_k = 1/K$

$$\text{L}_1 = \sum_{i=1}^{N}\sum_{k=1}^{K} w_{ik}\log\pi_k. \tag{7}$$

Likelihood term (L_2) measures the match between the samples and etalons

$$\text{L}_2 = \sum_{i=1}^{N}\sum_{k=1}^{K} w_{ik}\left[-(D-1)\log\beta_k - \frac{d_{ik}}{\beta_k}\right]. \tag{8}$$

Entropy term (L_3) encourages smoother soft-label assignments

$$\text{L}_3 = -\sum_{i=1}^{N}\sum_{k=1}^{K} w_{ik}\log w_{ik}. \tag{9}$$

Using the subgradient property of the L_1 distance, $\partial\|x_i - e_k\|_1/\partial e_k = \text{sign}(x_i - e_k)$, the gradient of the loss w.r.t. the etalon location is

$$\nabla_{e_k}\mathcal{L}' = -\frac{1}{N}\sum_{i=1}^{N} w_{ik}\frac{\text{sign}\left(x_i - e_k\right)}{\beta_k + \epsilon}. \tag{10}$$

Etalon locations are updated via gradient descent, encouraging each etalon to move toward regions of high sample density

$$e_k \leftarrow e_k - \eta\nabla_{e_k}\mathcal{L}'. \tag{11}$$

The gradient of the loss w.r.t. influence factor β_k is

$$\nabla_{\beta_k}\mathcal{L}' = -\frac{1}{N}\sum_{i=1}^{N} w_{ik}\left(\frac{D-1}{\beta_k + \epsilon} - \frac{d_{ik}}{(\beta_k + \epsilon)^2}\right). \tag{12}$$

The influence factors are updated similarly

$$\beta_k \leftarrow \beta_k - \eta\nabla_{\beta_k}\mathcal{L}'. \tag{13}$$

To gradually transition from broad exploration to fine-grained convergence, SWCA applies an exponential annealing schedule to the temperature parameter, with a minimum threshold set to 0.001 to maintain numerical stability and ensure progressively sharper soft assignments

$$\tau' \leftarrow \tau'\alpha. \tag{14}$$

4) Adaptive Etalon Resetting and Selection: To ensure that the condensed etalon set remains representative throughout training, SWCA incorporates a support-based evaluation and dynamic reset mechanism. The support of each etalon e_k is defined as the sum of soft-assignment weights $p_k = \sum_{i=1}^{N} w_{ik}$, which reflects the degree to which class samples depend on that etalon. A low support value indicates weak representativeness. After the warm-up period (10% of total iterations), the algorithm evaluates p_k every ten iterations. If $p_k < \theta$, the etalon is regarded as invalid and is reset by randomly selecting a class sample to replace its position while reinitializing its influence factor to $\beta_k = 1.0$ and its support to $p_k = 0$. This mechanism ensures that the etalon set consistently covers the core density regions of the class.

At the end of training, a second-stage screening is applied based on the learned influence factors. Only etalons satisfying $\beta_k > 0.4\cdot\bar{\beta}$, $\bar{\beta} = 1/K\sum_{k=1}^{K}\beta_k$ are retained, where $\bar{\beta}$ denotes the mean influence factor across all etalons. Etalons with excessively small scales—indicating a limited influence range—are removed to eliminate redundancy. The refined etalon set is then expressed as

$$\mathcal{E}^* = \left\{e_k^* | k \in \mathcal{K}^*\right\}. \tag{15}$$

The final output of SWCA consists of the core etalon set $\mathcal{E}^*$ and their corresponding influence factors β^*.

C. Open-Set Classification

Based on the discriminative feature space optimized by the HSCWM module and the category-specific core etalon sets produced by SWCA, this section introduces the complete HACL-DC detection framework. The goal is to achieve both accurate classification of known categories and reliable identification of unknown classes. The overall pipeline follows a feature metric–score computation–threshold decision paradigm, leveraging the hyperspherical feature structure and the weighted etalon representations to enhance robustness and accuracy in open-set hyperspectral recognition.

For a test sample (unseen during training), we first extract its initial hyperspectral feature vector $x_{test} \in \mathbb{R}^D$ via the same

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAIL INFORMATION ABOUT DATASETS

Indian Pines			WHU-Hi-LongKou			WHU-Hi-HanChuan		
No.	Class Name	Samples	No.	Class Name	Samples	No.	Class Name	Samples
1	Corn-notill	1428	1	Corn	34511	1	Strawberry	44735
2	Corn-mintill	830	2	Cotton	8374	2	Cowpea	22753
3	Grass-pasture	483	3	Broad-leaf soybean	63212	3	Soybean	10287
4	Hay-windrowed	478	4	Rice	11854	4	Sorghum	5353
5	Soybean-notill	972	5	Water	67056	5	Water spinach	1200
6	Soybean-mintill	2455	6	Unknown	19531	6	Watermelon	4533
7	Soybean-clean	593				7	Greens	5903
8	Woods	1265				8	Trees	17978
9	Unknown	1745				9	Grass	9469
						10	Bare soil	9116
						11	Water	75401
						12	Unknown	50802

feature extraction network used during training. To ensure consistency with the training procedure, the feature vector is L2-normalized, projecting it onto the unit hypersphere. For each known category $c \in \{1, 2, \ldots, C\}$, its core etalon set is denoted by $\mathcal{E}_c^* = \{e_{c,1}^*, e_{c,2}^*, \ldots, e_{c,K_c}^*\}$, with the corresponding learned influence coefficients $\beta_c^* = \{\beta_{c,1}^*, \beta_{c,2}^*, \ldots, \beta_{c,K_c}^*\}$.

To fully exploit the weighted representation characteristics of the etalons, we compute the similarity between the test sample and each category using a scale-adaptive distance metric. For category c, we compute the L_1 distance between the test sample and each etalon, and then adjust the distance using the learned influence coefficient. A larger $\beta_{c,k}^*$ indicates a broader influence range and lower local density, thus contributing a smaller weight to the distance. Conversely, etalons in dense regions contribute more

$$d_{c,k}^{\text{weighted}} = \frac{d_{c,k}}{\beta_{c,k}^* + \epsilon}. \tag{16}$$

In the hyperspherical space, ID samples tend to be close to at least one core etalon, while OOD samples remain distant from all etalons. Therefore, the minimum weighted distance provides the most sensitive indicator of category affiliation

$$S_c(\hat{x}_{\text{test}}) = \min_{k=1,2,\ldots,K_c} \left(1 - \frac{d_{c,k}^{\text{weighted}}}{\max_d + \epsilon}\right) \tag{17}$$

where max_d is the maximum weighted distance observed during training across all samples and etalons (precomputed). This term ensures that the similarity score $S_c(\hat{x}_{test}) \in [0, 1]$, with values closer to 1 indicating higher similarity to category c.

We compute similarity scores across all known categories and assign the test sample to the category with the highest score. A threshold γ is then applied to distinguish ID from OOD samples. If the maximum similarity score $S_c \geq \gamma$, the sample is regarded as an ID instance and classified into the corresponding category. Conversely, when $S_c < \gamma$, the sample is identified as an OOD instance and labeled as belonging to an unknown class

$$\hat{y}_i = \begin{cases} \arg\max_{c=1,2,\ldots,C} S_c(\hat{x}_{\text{test}}), & S_c^i \geq \gamma \\ C+1, & S_c^i < \gamma. \end{cases} \tag{18}$$

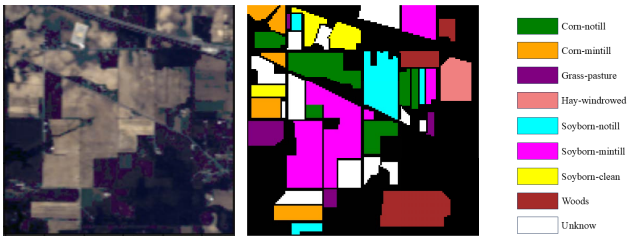


Fig. 5. Indian pines dataset. (a) Image data. (b) Ground truth. (c) Class name.

IV. EXPERIMENTAL RESULTS

A. Datasets Description

In this study, we evaluate the proposed HACL-DC framework using three hyperspectral datasets of different scales—Indian Pines (small), WHU-Hi-LongKou (medium), and WHU-Hi-HanChuan (large) [39]. Each dataset includes samples from unseen categories to facilitate OOD evaluation. A summary of the class labels and sample distributions for all datasets is provided in Table I.

1) *Indian Pines:* The Indian Pines dataset was acquired by NASA's Airborne Visible/Infrared Imaging Spectrometer over an agricultural area in northwestern Indiana. It has a spatial resolution of 20 m and a size of 145×145 pixels, with 224 spectral bands covering 400–2500 nm. After removing noisy and water-absorption bands, PCA is applied to reduce the data to 64 bands for model training. The dataset contains 16 land-cover classes with highly imbalanced samples. Following [40], classes with fewer than ten samples are treated as unknown, resulting in eight known classes. Class distributions and unknown samples are reported in Table I, and the corresponding spatial map is shown in Fig. 5.
2) *WHU-Hi-LongKou:* The WHU-Hi-LongKou dataset, a representative benchmark in the WHU-Hi series, was collected in Longkou, Shandong Province, China, using a Headwall hyperspectral imager. It has a spatial resolution of 0.463 m, an image size of 550×400 pixels, and 270 spectral bands spanning 400–1000 nm. The dataset includes nine land-cover classes, mainly crop types and economic forests [39]. To simulate realistic open-set conditions, categories such as roads, buildings,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

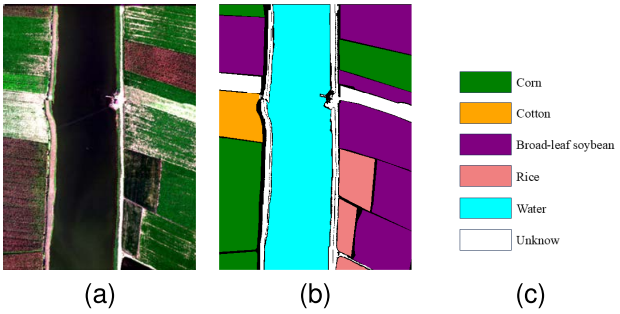


Fig. 6. WHU-Hi-LongKou. (a) Image data. (b) Ground truth. (c) Class name.



Fig. 7. WHU-Hi-HanChuan. (a) Image data. (b) Ground truth. (c) Class name.

and mixed weeds are treated as OOD samples. Class distributions are reported in Table I, and the spatial map is shown in Fig. 6.

3) *WHU-Hi-HanChuan:* The WHU-Hi-HanChuan dataset, a core benchmark in the WHU-Hi series, was collected in Hanchuan, Hubei Province, China, using a Headwall hyperspectral imager. It has a spatial resolution of 0.109 m, an image size of 1217 × 303 pixels, and 274 spectral bands covering 400–1000 nm. The dataset includes 16 land-cover classes, mainly agricultural crops. To simulate realistic open-set conditions, categories such as red and gray roofs, plastic surfaces, roads, and bright objects are treated as OOD samples, introducing partial spectral overlap with known classes. Classwise sample statistics are provided in Table I, and the spatial map is shown in Fig. 7.

To evaluate the classification performance of the proposed HACL-DC method under different levels of training-data availability, we set the number of training samples in each dataset to 50, 100, 150, and 200, respectively, while using all remaining samples as the test set. It is important to note that none of the training samples in any of the three datasets include data labeled as Unknown.

B. Experimental Implementation and Evaluation Metrics

HACL-DC consists of two modules, HSCWM and SWCA. In HSCWM, the intraclass and interclass slopes and intercepts are set to $S_b = 5$, $I_b = -1$, $S_w = 4$, $I_w = 1$. The temperature parameter is fixed at 0.1. We use the SGD optimizer with a learning rate of 0.01, momentum of 0.9, and a weight decay of 0.0001. Training is performed for 500 epochs with warm-up, cosine decay scheduling, and three step-decays by a factor of 0.1 at epochs 100, 300, and 400.

For SWCA, the value of K is set to 5 when the number of training samples is 50 or 100 and to 10 when the training sample size is 150 or 200. The initial temperature is set to 1.0 with a decay rate of 0.98. The confidence threshold θ is defined as 30% of the average support across all etalons. Gradient descent is used with a learning rate of 0.01 and a maximum of 300 iterations. A 13×13 neighborhood is used as the input patch, and all networks are trained with a batch size of 64. Data augmentation includes the addition of mild Gaussian noise, as well as random horizontal and vertical flips. All experiments are performed on the same hardware platform, which is equipped with an NVIDIA GTX 3090 GPU, an Intel Xeon Silver 4214 processor operating at 2.20 GHz, and 128 GB of RAM.

Classification performance is evaluated using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. For open-set evaluation, we use the accuracy of the unknown-class as the performance metric. The threshold is calibrated using only ID validation samples without accessing unknown data, following the standard OOD detection setting where OOD samples are unavailable during training. Specifically, the threshold is selected to achieve a 95% TPR on ID samples, corresponding to a 5% false ID rejection rate [41]. This calibration strategy provides a consistent tradeoff between maintaining ID classification performance and rejecting unknown samples.

C. Result Comparison and Analysis

To comprehensively evaluate the proposed method for open-set HSIC, we compare it with seven representative OOD detection approaches. The selected baselines are designed to cover diverse OOD detection paradigms and reflect the major research directions in this field. Specifically, we consider both postprocessing-based methods and training-based methods.

The postprocessing baselines focus on improving OOD detection performance without modifying the original training procedure, which represents a widely adopted strategy due to its simplicity and general applicability. This category includes rectified activation (ReAct) [42], activation shaping (ASH) [43], OOD detection with simplified hopfield energy (SHE) [44], and neural collapse inspired OOD detector (NCI) [45]. These methods cover representative postprocessing paradigms,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

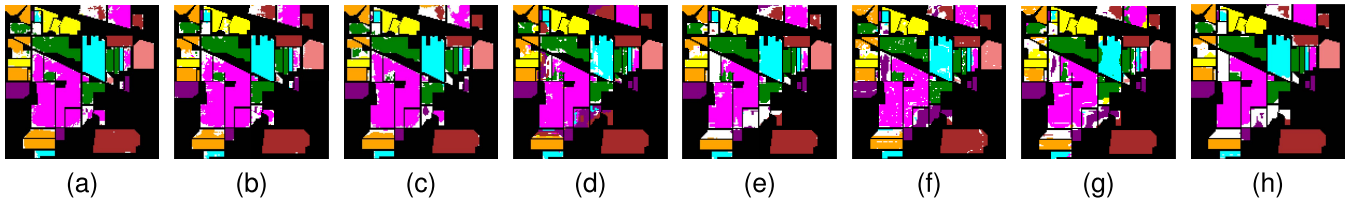


Fig. 8. Classification maps obtained by different methods on the indian pines dataset with 200 training samples. (a) ReAct. (b) ASH. (c) SHE. (d) NCI. (e) ISH. (f) PALM. (g) AscOOD. (h) HACL-DC.

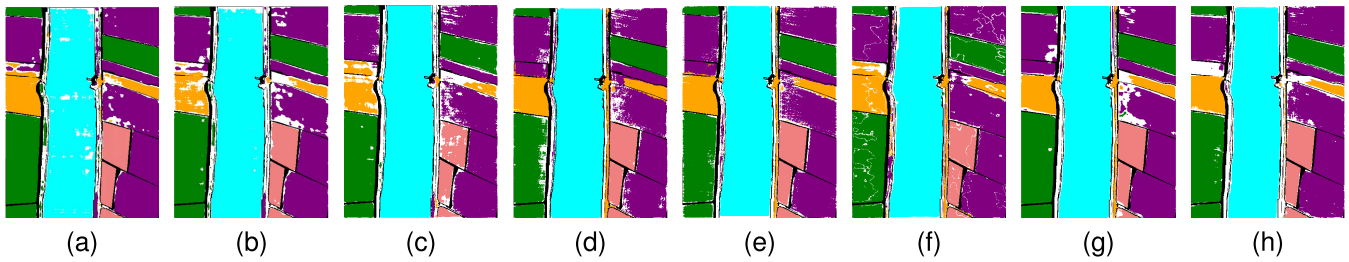


Fig. 9. Classification maps obtained by different methods on the WHU-Hi-LongKou dataset with 200 training samples. (a) ReAct. (b) ASH. (c) SHE. (d) NCI. (e) ISH. (f) PALM. (g) AscOOD. (h) HACL-DC.

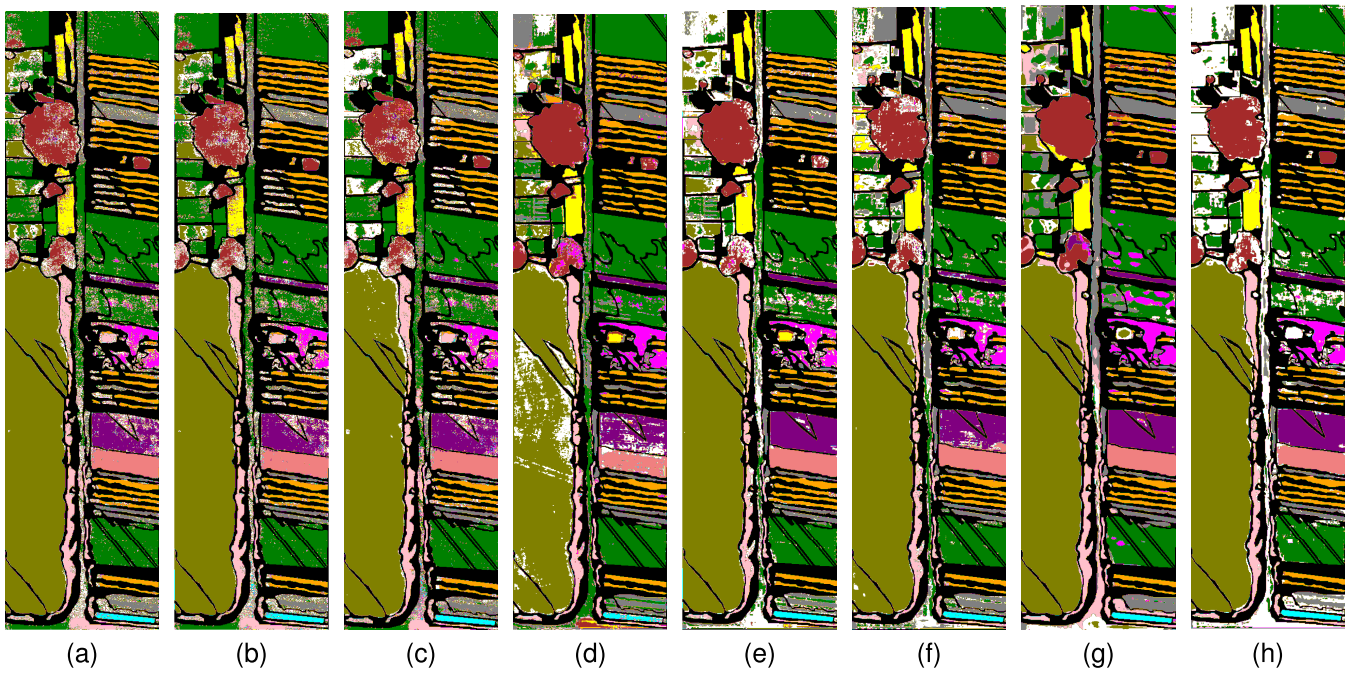


Fig. 10. Classification maps obtained by different methods on the WHU-Hi-HanChuan dataset with 200 training samples. (a) ReAct. (b) ASH. (c) SHE. (d) NCI. (e) ISH. (f) PALM. (g) AscOOD. (h) HACL-DC.

including activation manipulation, energy-based scoring, and neural collapse-inspired decision strategies.

The training-based baselines aim to enhance OOD detection by learning more discriminative feature representations during model optimization. We select intermediate tensor SHaping (ISH) [46], PrototypicAl learning with a Mixture of prototypes (PALMs) [41], and approach to Spurious, fine-grained, and conventional OOD detection (AscOOD) [47]. These methods represent recent advances in feature transformation, prototype-based learning, and unified OOD detection frameworks, respectively. Since the proposed method also improves OOD discrimination through representation optimization and prototype-based modeling, these methods constitute appropriate and representative baselines for comparison.

All models are trained and tested using the parameter settings recommended in their original publications. Tables II–VII summarize the quantitative results of all methods on the three datasets. Specifically, Tables II, IV, and VI report the unknown-class accuracy (Unknow), AA, OA, and Kappa coefficient for different training sample sizes, while Tables III, V, and VII summarize the ID classification results with 50 training samples. Figs. 8–10 present the classification maps for each method when the number of training samples is 200.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

TABLE II

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE INDIAN PINES DATASET (%)

Sample size		ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
50	Uknow	10.81±2.84	10.87±2.54	12.28±2.91	16.37±9.84	15.32±3.39	19.45±5.08	12.05±4.53	**29.34±2.89**
	AA	67.88±2.09	65.44±1.69	64.77±1.41	72.44±5.23	76.99±3.20	85.26±0.76	74.24±4.92	**86.83±0.26**
	OA	57.37±3.12	54.09±1.80	54.27±1.64	65.68±2.62	68.61±3.13	77.70±0.71	65.75±6.09	**80.59±0.34**
	Kappa	50.94±3.32	47.44±1.85	47.61±1.76	60.72±3.18	64.05±3.66	74.41±0.81	60.93±6.70	**77.58±0.33**
100	Unknow	21.66±4.74	12.10±1.45	16.37± 6.54	5.31±3.43	24.53±4.23	32.09± 5.40	39.37± 13.95	**54.90±8.02**
	AA	85.15±0.73	84.79±0.27	85.08± 0.82	83.13±1.28	86.86±0.20	88.28± 0.28	87.67± 1.35	**91.11±1.03**
	OA	76.99±1.39	76.37±0.65	77.04± 1.43	73.93±0.54	79.15±0.35	81.56± 0.52	81.58± 2.72	**86.72±2.15**
	Kappa	73.21±1.51	72.41±0.71	73.17± 1.60	70.06±0.67	75.90±0.41	78.67± 0.56	78.66± 3.02	**84.51±2.47**
150	Unknow	35.09±3.81	28.06±4.35	26.12±3.70	23.23± 10.79	31.44±8.16	38.85±9.40	53.96±3.68	**74.40±2.88**
	AA	89.42±0.46	88.70±0.41	88.21±0.41	83.12± 2.42	87.92±0.98	89.53±1.01	89.76±2.13	**93.78±0.25**
	OA	82.92±0.68	81.59±0.90	80.94±0.93	77.06± 1.47	80.70±1.57	83.59±1.82	84.87±2.26	**90.87±0.54**
	Kappa	79.91±0.78	78.29±1.06	77.53±1.09	73.36± 1.57	77.56±1.73	80.86±2.11	82.37±2.56	**89.24±0.63**
200	Unknow	42.35±5.36	42.32±5.13	42.28±4.71	22.71± 12.66	58.91±5.81	45.83±16.75	66.59±11.09	**81.64±1.27**
	AA	89.37±1.48	89.90±1.46	90.08±0.32	87.57± 1.27	92.08±0.90	89.52±1.52	92.05±3.15	**94.47±0.21**
	OA	84.28±1.13	84.19±1.06	84.08±0.90	78.09± 2.34	87.02±1.53	84.67±3.36	88.09±4.21	**92.31±0.27**
	Kappa	81.30±1.33	81.22±1.28	81.08±1.07	74.56± 2.64	84.72±1.77	82.05±3.83	85.97±4.90	**90.88±0.29**

TABLE III

ID CLASSIFICATION RESULTS ON INDIAN PINES WITH 50 TRAINING SAMPLES (%)

ClassNo.	ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
1	28.24±5.14	38.62±3.37	38.72± 4.21	38.72±7.32	71.49±6.50	86.84± 1.26	68.43±6.83	**89.24±0.74**
2	85.77±4.10	75.69±4.50	72.56± 12.86	72.56±4.66	82.71±1.05	95.64± 1.99	77.82± 4.80	**95.85±0.97**
3	94.18±0.83	92.98±2.33	89.75± 3.28	89.75±36.92	95.55±4.99	**99.62± 0.29**	95.23±1.94	99.15±0.44
4	99.53±0.72	99.53±0.82	98.46± 1.73	98.46±1.32	99.21±1.12	99.61± 0.29	99.92±0.11	**100.00±0.00**
5	67.51±6.60	66.36±17.98	62.10± 17.52	62.10±9.87	77.27±11.21	90.74± 3.33	76.03±8.05	**90.93±3.48**
6	57.19±11.57	43.06±7.12	47.18± 3.95	47.18±3.07	71.56±4.78	81.57± 3.69	68.07±9.97	**86.32±1.00**
7	69.24±10.39	63.61±9.07	67.40± 7.80	67.40±4.92	85.79±5.16	**95.40± 1.45**	79.62±5.55	95.21±2.08
8	98.44±0.86	98.26±1.39	94.47± 4.90	94.47±14.61	93.98±1.01	**98.46± 0.37**	90.95±2.35	95.45±0.78

TABLE IV

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE WHU-HI-LONGKOU DATASET (%)

Sample size		ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
50	Unknow	35.86± 20.94	29.97±11.75	27.68±11.63	15.59±13.63	47.33±3.30	26.23± 7.62	30.97±5.89	**63.05±4.93**
	AA	84.43± 4.21	83.59±3.15	83.24±3.05	81.15±0.78	86.74±0.62	83.18± 0.91	82.89±1.46	**89.57±0.92**
	OA	86.81± 3.52	85.32±3.40	85.11±3.34	86.28±1.62	89.98±0.51	88.10± 0.75	87.26±1.38	**91.96±0.48**
	Kappa	82.71± 4.61	80.79±4.46	80.53±4.35	81.96±2.07	86.84±0.66	84.24± 1.00	83.33±1.76	**89.41±0.64**
100	Unknow	65.42±6.58	72.34±5.05	74.13±4.55	32.32±6.45	51.17±2.13	27.79± 6.52	43.40±6.34	**76.94±3.67**
	AA	90.48±1.35	91.07±1.49	90.74±1.74	85.48±1.45	88.33±0.42	83.99± 1.57	85.75±1.06	**91.62±0.75**
	OA	92.16±0.63	92.82±0.49	92.99±0.44	88.64±0.69	90.66±0.24	88.30± 0.70	89.26±1.05	**93.32±0.38**
	Kappa	89.71±0.84	90.56±0.64	90.77±0.60	85.04±0.95	87.73±0.33	84.55± 0.94	85.91±1.35	**91.19±0.49**
150	Unknow	67.30±5.90	73.51±5.83	74.08±3.51	20.22±9.17	55.64±3.46	31.94± 2.85	46.15±5.03	**80.17±0.80**
	AA	90.11±2.36	91.92±0.41	91.68±0.71	80.99±4.92	89.38±0.36	84.14± 0.35	87.14±0.79	**92.69±0.48**
	OA	92.34±0.57	92.94±0.56	92.99±0.34	87.25±0.78	91.19±0.29	88.84± 0.30	90.17±0.47	**93.53±0.05**
	Kappa	89.93±0.75	90.73±0.73	90.80±0.44	83.15±1.08	88.41±0.39	85.24± 0.39	87.08±0.62	**91.50±0.09**
200	Unknow	67.84±6.02	65.20±4.86	73.93±1.68	36.03±2.22	53.02±2.63	38.84± 8.71	54.29±3.38	**84.39±0.12**
	AA	91.67±0.92	91.09±0.28	90.43±1.18	85.54±0.49	88.55±0.41	85.43± 1.49	86.52±3.11	**92.97±0.08**
	OA	92.39±0.58	92.14±0.47	92.97±0.15	88.55±0.30	90.81±0.24	89.51± 0.77	89.44±2.57	**94.05±0.10**
	Kappa	90.01±0.75	89.70±0.60	90.76±0.19	84.94±0.37	87.91±0.33	86.16± 1.08	86.16±3.31	**92.17±0.13**

TABLE V

ID CLASSIFICATION RESULTS ON WHU-HI-LONGKOU WITH 50 TRAINING SAMPLES (%)

ClassNo.	ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
1	98.04±1.43	98.78±1.11	98.85±0.94	**99.12±0.31**	98.09±0.69	95.78± 3.49	97.04±1.14	95.82±0.40
2	94.83±3.04	98.19±1.00	**98.34±1.28**	93.13±5.21	92.41±2.04	91.80± 1.90	93.85±1.07	92.92±0.46
3	80.38±6.64	75.81±7.05	75.73±7.29	86.55±5.96	89.73±1.11	**95.65± 1.42**	87.59±3.26	94.87±0.94
4	98.58±0.82	**99.40±0.23**	99.39±0.24	94.92±7.18	95.73±3.69	96.57± 0.78	91.10±0.73	96.18±0.92
5	98.87±1.20	99.42±0.65	**99.48±0.65**	97.60±2.51	97.11±0.46	93.04± 3.37	96.80±1.64	97.59±0.94

Beyond classification performance, we further evaluate the computational efficiency of different methods by comparing their training and inference time on the WHU-Hi-HanChuan dataset, which is the largest and most challenging dataset among the three datasets in terms of spatial size, spectral dimension, and class number. The computational time comparison results are summarized in Table VIII.

The experimental results indicate that several postprocessing approaches, including ReAct, ASH, SHE, and NCI evaluated in this study, can achieve relatively high OA but generally suffer from poor unknown-class detection performance. A limitation shared by these methods is that their performance gains are strongly bounded by the quality of the pretrained base classifier. By design, such approaches attempt to identify unknown samples through post hoc calibration of model outputs, for example, activation values, entropy, or confidence scores. However, when the underlying classifier fails to learn sufficiently discriminative features in complex scenarios, the effectiveness of postprocessing becomes inherently limited.

Specifically, ReAct and ASH improve open-set detection mainly by reshaping the output distribution, yet they perform noticeably worse on datasets with higher class complexity,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET (%)

Sample size		ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
50	Unknow	8.84±1.85	7.95±3.79	7.64±3.69	5.85±2.96	10.31±2.27	13.30± 7.21	5.86±5.47	**25.59±5.15**
	AA	64.52±5.77	64.10±8.64	63.99±8.45	72.79±0.60	75.25±0.38	79.68± 0.32	73.24±2.81	**82.45±0.91**
	OA	65.16±4.66	64.34±5.48	64.20±5.79	69.81±0.72	71.30±0.51	73.04± 2.48	69.97±2.34	**77.46±0.49**
	Kappa	58.45±5.22	57.50±6.33	57.37±6.64	63.74±0.94	65.56±0.64	68.10± 2.65	63.98±2.84	**73.05±0.49**
100	Unknow	10.82±3.72	7.44±2.47	6.54±3.97	11.69±5.64	19.89±7.01	27.52± 0.83	20.57±6.69	**41.12±5.25**
	AA	69.30±1.38	69.74±0.84	68.87±0.80	76.11±5.33	80.32±2.77	85.33± 0.28	81.66±2.64	**87.56±0.92**
	OA	67.70±1.08	68.39±0.85	67.49±0.67	71.73±2.66	75.54±2.51	79.43± 0.09	75.51±1.79	**82.85±1.18**
	Kappa	61.26±1.17	61.95±1.01	61.00±0.83	66.52±3.17	70.82±2.93	75.30± 0.07	70.67±2.09	**79.38±1.46**
150	Unknow	13.87±1.43	12.13±2.32	10.61±3.12	14.30±6.94	27.16±9.27	35.29± 4.49	34.00±0.53	**52.21±4.01**
	AA	79.12±1.56	79.35±1.34	78.87±1.57	82.84±2.02	84.82±1.09	87.59± 0.74	86.53±2.12	**88.80±0.02**
	OA	74.01±1.26	74.12±1.34	73.55±1.44	74.90±0.92	78.98±1.94	81.39± 0.99	81.21±1.27	**84.80±0.44**
	Kappa	68.76±1.52	68.81±1.64	68.22±1.80	70.19±1.06	74.78±2.28	77.75± 1.17	77.37±1.52	**81.66±0.51**
200	Unknow	16.21±1.05	14.45±1.68	14.35±3.98	20.09±8.36	35.10±7.95	36.08± 1.60	40.53±2.77	**61.91±1.39**
	AA	82.41±0.82	82.31±0.99	81.91±1.03	84.73±1.40	86.43±1.05	87.40± 0.61	88.75±0.58	**91.31±0.08**
	OA	75.88±0.93	76.26±0.59	75.84±0.71	76.97±1.66	81.57±1.69	81.89± 0.21	81.65±1.40	**88.02±0.34**
	Kappa	71.04±1.06	71.39±0.68	71.01±0.89	72.59±1.98	77.81±2.00	78.32± 0.27	78.08±1.59	**85.53±0.40**

TABLE VII

ID CLASSIFICATION RESULTS ON WHU-HI-HANCHUAN WITH 50 TRAINING SAMPLES (%)

ClassNo.	ReAct	ASH	SHE	NCI	ISH	PALM	AscOOD	HACL-DC
1	75.15±15.48	73.65±11.25	73.61±13.98	90.94±1.35	88.49±2.41	**92.28± 2.14**	90.01±1.59	90.57±0.72
2	77.83±5.74	75.79±3.13	75.70±4.06	78.63±3.87	76.67±3.89	**85.34± 3.30**	74.25±5.58	84.95±1.24
3	66.14±10.83	71.10±12.72	69.26±11.77	85.44±7.04	88.20±4.40	92.53± 0.78	82.18±6.51	**94.41±1.52**
4	**97.36±1.29**	96.60±1.06	95.99±1.40	96.31±1.31	94.66±1.43	94.91± 1.37	93.60±1.58	96.25±0.53
5	83.58±10.81	87.18±10.17	87.22±8.32	97.54±1.31	97.04±1.69	97.91± 0.44	99.36±0.11	**99.88±0.16**
6	31.81±15.66	27.45±21.07	30.32±21.71	58.58±2.94	59.94±1.96	**80.13± 5.20**	57.03±1.29	76.50±4.33
7	88.59±9.36	87.15±8.26	83.73±9.13	85.41±5.09	91.75±2.89	93.00± 0.56	**93.42±0.50**	89.89±4.76
8	43.27±8.71	38.32±21.34	38.75±21.79	63.07±6.51	61.67±3.31	66.82± 1.14	65.43±2.92	**70.62±1.09**
9	51.55±20.63	48.55±19.77	51.31±19.10	55.91±8.47	67.25±5.70	71.45± 3.98	47.19±9.55	**83.00±4.77**
10	50.65±9.70	56.07±13.31	55.13±13.53	58.20±5.43	68.46±1.11	75.47± 4.51	71.70±4.16	**79.49±4.94**
11	**99.51±0.37**	99.43±0.25	99.27±0.24	97.54±1.97	98.50±0.18	93.06± 7.05	98.90±0.68	98.20±0.47

TABLE VIII

COMPUTATIONAL TIME COMPARISON OF DIFFERENT METHODS ON THE WHU-HI-HANCHUAN DATASET

Method	Training Time (s)	Inference Time (ms)
ReAct	–	3133
ASH	–	3815
SHE	–	5556
NCI	–	4458
ISH	35	3669
PALM	688	3422
AscOOD	2452	6564
HACL-DC (Ours)	610 (HSCWM) 29 (SWCA)	4080

such as WHU-Hi-HanChuan. On this dataset, both OA and unknown-class accuracy are substantially lower than those achieved by training-based methods, indicating that output-level calibration alone cannot address deficiencies in feature discriminability. SHE and NCI, which rely on information-theoretic criteria such as entropy and consensus, exhibit pronounced instability. For instance, NCI shows large fluctuations in unknown-class accuracy on the Indian Pines dataset across different training sizes, with standard deviations reaching ±12.66, and demonstrates no consistent performance trend on WHU-Hi-LongKou as the number of training samples increases. These observations highlight a key weakness of postprocessing methods, namely their high sensitivity to data distributions and model states, resulting in limited robustness.

Training-based approaches, including ISH, PALM, and AscOOD, can optimize feature representations through end-to-end learning, but they face a different set of challenges. Their common limitation lies in the difficulty of balancing compact known-class feature clustering with effective rejection of unknown samples. ISH, in particular, shows insufficient open-set recognition capability. Although it attains reasonable closed-set accuracy in some settings, its unknown-class accuracy remains consistently low across all datasets. Even with 200 training samples on WHU-Hi-HanChuan, ISH achieves an unknown-class accuracy of only 35.10%, suggesting that its training objective primarily focuses on minimizing closed-set classification error without encouraging sensitivity to OOD samples.

PALM and AscOOD explicitly incorporate open-set recognition mechanisms, yet fail to achieve a satisfactory balance between performance and stability. PALM achieves relatively high closed-set OA on Indian Pines, but its unknown-class accuracy is markedly lower than that of HACL-DC and tends to saturate as the training sample size increases. Among the compared methods, AscOOD shows relatively strong overall performance but remains clearly inferior to the proposed HACL-DC in both classification and open-set detection, particularly on challenging datasets. On the WHU-Hi-HanChuan dataset with 200 training samples, AscOOD achieves an OA of 81.65% and unknown-class accuracy of 40.53%, compared with 88.02% and 61.91% for HACL-DC. These differences highlight fundamental limitations in AscOOD's feature learning and decision boundary modeling, as well as limited intraclass consistency, with noticeably lower accuracy on several individual classes.

In contrast, the proposed HACL-DC method demonstrates clear advantages across all datasets, achieving a favorable balance between closed-set classification and open-set detection. On the Indian Pines dataset with 200 training

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

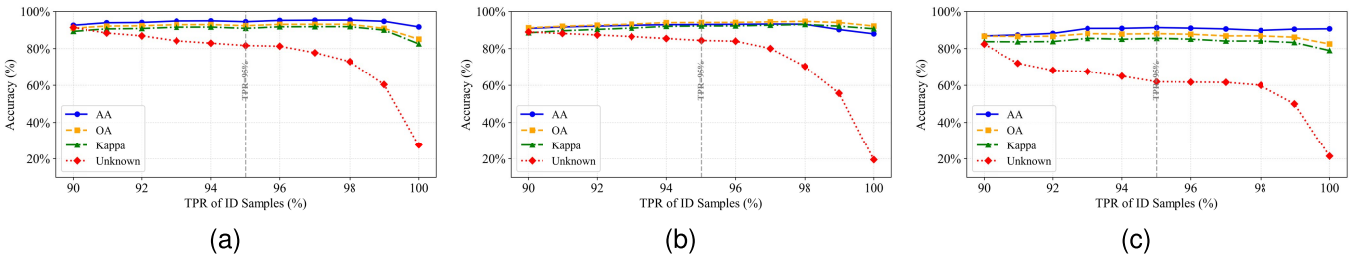


Fig. 11. Threshold sensitivity analysis of HACL-DC under different ID TPR constraints on three datasets. (a) Indian pines. (b) WHU-Hi-LongKou. (c) WHU-Hi-HanChuan.

samples, HACL-DC attains an OA of 92.31% together with an unknown-class accuracy of 81.64%, substantially outperforming all competing methods. On the most challenging WHU-Hi-HanChuan dataset, HACL-DC achieves an OA of 88.02% and an unknown-class accuracy of 61.91%, maintaining leading performance on both metrics. These results indicate that HACL-DC improves open-set detection without compromising closed-set classification performance.

As shown in Table V, in the ID classification task on the WHU-Hi-LongKou dataset, postprocessing methods such as ReAct, ASH, and SHE avoid model training and directly compare test samples with prior spectral signatures of known classes. Several classes in this dataset, including Classes 1, 2, and 5, exhibit simple and representative spectral characteristics, allowing effective classification by measuring distances to class centers without relying on deep feature extraction or classifier training. In contrast, training-based methods such as PALM and HACL-DC must simultaneously optimize feature generalization and open-set detection during training, resulting in more complex learning objectives. This tradeoff may slightly reduce classification accuracy on some simple classes compared with direct spectral matching methods, leading to minor differences in ID performance. Nevertheless, even when the per-class accuracy is marginally lower, HACL-DC consistently achieves substantially higher OA than other competing methods.

The classification maps in Figs. 8–10 further demonstrate the advantages of HACL-DC. Compared with other methods, HACL-DC yields clearer class boundaries while maintaining high closed-set classification accuracy. Although some approaches achieve competitive OOD detection in quantitative metrics, they often misclassify many ID samples as unknown. In regions where ID and OOD samples coexist, HACL-DC provides more reliable separation, an effect that is particularly evident on the Indian Pines dataset where the proportion of OOD samples is relatively low. Even in this challenging setting, HACL-DC produces classification results that closely match the ground truth while preserving effective unknown-class detection. Overall, its classification maps exhibit sharper interclass boundaries and reduced misclassification noise. These results indicate that the contrastive similarity–based feature space learned by HACL-DC better captures intraclass similarity and interclass discrepancy, providing a stronger foundation for open-set HSIC.

In addition to classification effectiveness, computational efficiency is another important factor for evaluating open-set HSIC methods. As shown in Table VIII, the computational time varies considerably among different methods due to their different optimization strategies and inference procedures. For postprocessing-based methods, including ReAct, ASH, SHE, and NCI, no additional training time is required because they do not modify the original classifier training process. Their computational overhead is only introduced during inference, such as activation adjustment, energy calculation, or score calibration. In contrast, training-based methods require additional optimization to learn OOD-aware feature representations, resulting in extra computational costs. For example, ISH introduces feature transformation through scale-factor optimization during backpropagation for only 20 epochs, leading to relatively low training overhead. PALM and AscOOD further optimize prototype representations or unified OOD objectives, which results in additional training costs. For the proposed HACL-DC, the training time mainly consists of HSCWM-based feature learning and SWCA-based etalon optimization. HSCWM introduces only similarity weighting without additional network structures, while SWCA performs iterative etalon updates only during training. After training, inference only requires feature extraction and distance calculation between test samples and etalons. Therefore, HACL-DC achieves a favorable tradeoff between computational cost and open-set recognition performance.

D. Threshold Sensitivity Analysis

The decision threshold is an important factor in open-set HSIC, as it controls the tradeoff between retaining ID samples and rejecting unknown samples. In this work, the threshold is determined during training by specifying a target TPR for the validation ID samples, and a 95% TPR is adopted as the default setting. To investigate whether the performance of HACL-DC strongly depends on this specific threshold selection, we conduct a sensitivity analysis by varying the target TPR of validation ID samples from 90% to 100%. The results on the three datasets are shown in Fig. 11.

When the target ID TPR is increased, the decision threshold becomes more conservative, resulting in a stricter acceptance criterion for ID samples and a stronger rejection tendency toward uncertain samples. Conversely, lower TPR constraints relax the decision boundary and may increase the risk of accepting unknown samples. As shown in Fig. 11, HACL-DC exhibits relatively stable performance within a broad range of intermediate TPR settings, while only the extreme cases

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION RESULTS OF THE HSCWM AND SWCA MODULES IN HACL-DC (%)

Dataset	Baseline	HSCWM	SWCA	AA	OA	Kappa	Unknow
Indian Pines	✓			91.96±1.72	88.43±1.16	86.35±1.34	65.79±2.53
	✓	✓		93.35±1.00	90.82±1.06	89.13±1.26	77.69±3.49
	✓		✓	93.52±0.14	89.87±0.44	88.01±0.51	69.78±1.93
	✓	✓	✓	**94.47±0.21**	**92.31±0.27**	**90.88±0.29**	**81.64±1.27**
WHU-Hi-LongKou	✓			90.46±0.64	92.39±0.27	89.98±0.35	67.73±2.67
	✓	✓		92.46±0.50	93.36±0.50	91.28±0.66	79.32±3.57
	✓		✓	90.84±0.47	92.60±0.20	90.22±0.30	69.70±2.64
	✓	✓	✓	**92.97±0.08**	**94.05±0.10**	**92.17±0.13**	**84.39±0.12**
WHU-Hi-HanChuan	✓			88.78±0.88	83.07±2.81	79.70±3.23	44.03±8.38
	✓	✓		90.27±0.04	86.31±0.12	83.50±0.17	53.85±0.43
	✓		✓	89.90±1.15	86.36±2.51	83.70±3.03	55.16±11.14
	✓	✓	✓	**91.31±0.08**	**88.02±0.34**	**85.53±0.40**	**61.91±1.39**

with overly low or high TPR constraints show noticeable performance degradation. This indicates that the effectiveness of HACL-DC does not rely on a specifically optimized threshold value. Instead, its robust open-set recognition capability mainly benefits from the discriminative feature representations learned by HSCWM and the etalon-based decision strategy introduced by SWCA. The default 95% TPR setting provides a balanced tradeoff between retaining ID samples and rejecting unknown samples. Moreover, the performance remains relatively stable when the TPR varies within a range of approximately ±3% around this value, indicating that the proposed method does not rely on a precisely tuned threshold.

E. Ablation Study

In this section, we investigate the effectiveness of the proposed HSCWM and SWCA modules. Table IX summarizes the classification results of different model variants on the datasets when the number of training samples is set to 200. The baseline model is trained and evaluated using standard contrastive learning, with the decision threshold determined during training to achieve a TPR of 95% for ID samples.

The experimental results show that incorporating the HSCWM module alone yields consistent and significant improvements over the baseline across all three datasets. The AA, OA, and the Kappa coefficient all increase, together with notable gains in OOD detection performance. On the Indian Pines dataset, for instance, OA improves from 88.43% to 90.82%, while unknown-class accuracy rises from 65.79% to 77.69%. These results demonstrate that HSCWM enhances feature discriminability by adaptively assigning higher weights to hard intraclass positive pairs and interclass negative pairs. By concentrating the training process on samples that are most critical for promoting intraclass compactness and interclass separability, HSCWM mitigates the loose feature clustering caused by uniform weighting in conventional contrastive learning. As a result, it provides a stronger feature foundation for both ID classification and OOD detection.

Similarly, introducing the SWCA module alone leads to consistent performance improvements across all datasets. By incorporating soft-label assignment, dual-parameter gradient optimization, and adaptive etalon resetting, SWCA constructs compact and representative etalon sets that capture fine-grained class distributions and overcome the limitations of single-center class modeling. However, in the absence of HSCWM, less compact feature representations and random etalon initialization can lead to slightly higher performance variance on some datasets, such as the unknown-class accuracy on WHU-Hi-HanChuan.

When HSCWM and SWCA are combined, the model achieves substantially greater gains than either module alone, reflecting strong complementarity between feature optimization and class representation. On the Indian Pines dataset, OA increases from 89.87% with a single module to 92.31% with both modules, and unknown-class accuracy rises from 69.78% to 81.64%. On the WHU-Hi-LongKou and WHU-Hi-HanChuan datasets, the synergistic effect of the modules also exhibits a consistent trend of performance improvement, with the unknown-class accuracy increasing by 14.69% and 8.06%, respectively. This synergy arises because HSCWM produces a compact and well-separated feature space that facilitates accurate etalon construction by SWCA, while the weighted etalon-based distance measurement further amplifies the distinction between ID and OOD samples.

Moreover, the magnitude of performance gains varies across datasets. Larger improvements are achieved on datasets characterized by high spectral variability and dispersed sample distributions, such as Indian Pines. In contrast, the OA improvements are more moderate, yet remain statistically significant, on datasets with more compact spectral characteristics, such as WHU-Hi-LongKou. Finally, the low standard deviations observed for all metrics in the complete framework, for example an OA standard deviation of only 0.34 on WHU-Hi-HanChuan, confirm that the joint use of HSCWM and SWCA preserves training stability and ensures robust performance of the proposed HACL-DC framework.

Although the above experiments verify the effectiveness of the proposed HSCWM and SWCA modules, the contribution of individual design choices within each module has not been fully investigated. Therefore, we further conduct fine-grained ablation experiments by replacing or removing specific components, including the similarity weighting function, similarity transformation, soft-label assignment, and etalon reset strategy. The results are summarized in Table X.

All variants achieve lower performance than the complete HACL-DC model, indicating that each component contributes to the overall effectiveness of the proposed framework. For HSCWM, replacing the sigmoid-based weighting function with ReLU weighting leads to a consistent decline in unknown-class accuracy across all datasets. This suggests that the smooth nonlinear weighting mechanism of the sigmoid is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

TABLE X
FINE-GRAINED ABLATION STUDY OF DIFFERENT DESIGN CHOICES IN HACL-DC (%)

Dataset	Variant	OA	AA	Kappa	Unknown
Indian Pines	w/ ReLU-based weighting	91.94	95.11	90.24	79.09
	w/o similarity transformation	92.08	94.16	90.33	80.41
	w/o soft-label assignment	91.65	94.14	89.97	78.83
	w/o adaptive reset strategy	91.99	94.26	90.43	80.94
WHU-Hi-LongKou	w/ ReLU-based weighting	92.31	91.03	89.99	73.88
	w/o similarity transformation	93.19	91.96	90.64	80.78
	w/o soft-label assignment	92.64	90.98	90.32	76.37
	w/o adaptive reset strategy	93.88	92.78	92.00	83.40
WHU-Hi-HanChuan	w/ ReLU-based weighting	86.61	89.51	83.77	57.14
	w/o similarity transformation	87.16	91.07	84.54	60.96
	w/o soft-label assignment	87.40	90.35	84.11	58.94
	w/o adaptive reset strategy	87.64	91.09	84.35	60.59

more suitable for characterizing the continuous hardness distribution of sample pairs. In contrast, ReLU mainly performs one-sided activation according to a fixed threshold, which may limit its ability to assign differentiated optimization importance to samples with different difficulty levels. Moreover, removing the similarity transformation before sigmoid weighting also results in performance degradation. This indicates that the scaling and shifting operations are beneficial for adapting the similarity distribution to the effective operating range of the sigmoid function. Since cosine similarities are usually concentrated within a relatively narrow interval, directly applying sigmoid weighting may weaken the distinction between hard and easy samples, while the transformation provides additional flexibility to emphasize informative hard positive and hard negative pairs.

For SWCA, removing the soft-label assignment results in a consistent decrease in performance, especially for unknown-class recognition. This observation indicates that hard class assignment may be insufficient for hyperspectral images, where intraclass variations and interclass similarities can lead to ambiguous feature distributions. By introducing soft labels, SWCA can preserve more fine-grained class membership information during etalon optimization, allowing the condensed representations to better characterize the distribution of known classes. Furthermore, removing the adaptive reset strategy also causes performance degradation, suggesting that periodically updating etalons helps avoid suboptimal representations during optimization. Without this strategy, the learned etalons may become less representative due to the dynamic changes of feature distributions, thereby affecting the robustness of unknown-class detection.

Overall, these fine-grained ablation results suggest that the individual components of HACL-DC provide complementary contributions to feature discrimination and unknown-class rejection, supporting the effectiveness of the proposed design choices for open-set HSIC.

F. Hyperparameter Sensitivity Analysis

The proposed HACL-DC involves several hyperparameters that influence the feature-learning and class-representation quality. Among them, the number of etalons K in the SWCA module is a critical structural hyperparameter, as it directly determines the representation capacity of each known class. Etalons serve as core class representatives in the SWCA module to model the distribution of known classes. Therefore, the value of K affects not only the completeness of class distribution modeling but also the computational and memory costs of the model, leading to a tradeoff between known-class classification accuracy and unknown-class detection capability. When K is too small, the model may fail to capture the fine-grained and multimodal distributions of hyperspectral data. Conversely, an excessively large K introduces redundant computation and may increase the risk of overfitting. Therefore, we conduct a sensitivity analysis of K to investigate its influence on both ID classification and OOD detection performance.

Other parameters, such as the slope and intercept parameters S and I in HSCWM, are not treated as independent empirical hyperparameters. Instead, they are designed according to the characteristics of the sigmoid-based weighting function and the optimization objectives of positive and negative sample pairs. Specifically, the transformed similarity range is adjusted to ensure that hard positive pairs and hard negative pairs receive larger weights, while easy pairs are assigned lower optimization importance. Therefore, these parameters are determined based on the desired weighting behavior rather than extensive parameter searching.

To investigate the optimal range of K and its impact on performance, we conduct systematic experiments on three hyperspectral datasets, analyzing the variation in ID classification accuracy and OOD detection performance under different values of K. Since etalons are generated from training samples, the number of training samples is fixed at 200 to support evaluation with a larger range of K. The tested values of K range from 3 to 30, covering the commonly used range of etalon numbers in hyperspectral classification tasks. Consistent with previous experiments, OA, AA, and the Kappa coefficient are adopted to evaluate ID classification performance. The results are shown in Fig. 12. In addition to accuracy metrics, inference time under different values of K is also reported to analyze the tradeoff between performance and computational cost.

Through analysis of the results, it can be observed that OOD detection on the Indian Pines is more sensitive to K than ID classification, although the variation remains well controlled. The optimal performance is achieved at $K = 15$, where the etalon set simultaneously preserves high ID classification accuracy and maximizes feature separability between ID and OOD samples. For the WHU-Hi-LongKou dataset, the optimal results are achieved when K lies between 8 and 18. This indicates that the class distributions contain well-defined core structures that are effectively captured within this range. OOD detection shows even more sensitive to K, with peak performance occurring in the same range. In particular, moderate values of K provide a clear separation between ID and OOD samples, whereas excessive etalons blur this boundary and lead to a sharp decline in detection performance. For the WHU-Hi-HanChuan dataset, ID classification performance shows localized peaks while remaining stable overall. The optimal performance is achieved at $K = 10$, where the etalon set most accurately captures the core class distributions. Deviations from this value introduce either insufficient coverage or redundancy, leading to noticeable reductions in AA and Kappa.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

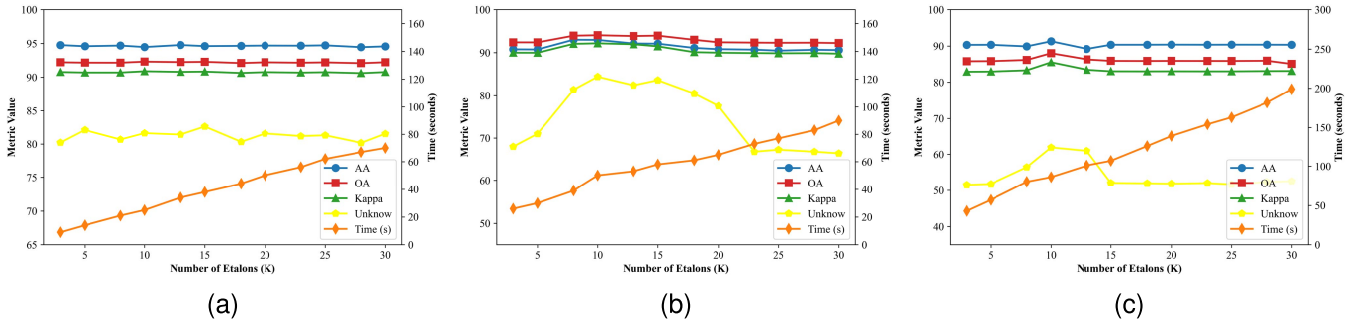


Fig. 12. Effect of the etalon number K on classification and OOD detection performance. (a) Results on the indian pines dataset. (b) Results on the WHU-Hi-LongKou dataset. (c) Results on the WHU-Hi-HanChuan dataset. Each subfigure illustrates the variation of OA, AA, Kappa coefficient, unknown-class accuracy, and inference time with different values of K.

Overall, the results indicate that ID classification is robust to a wide range of K values when sufficient training samples are available, whereas OOD detection is more sensitive to the choice of K. Across all datasets, optimal performance is achieved when K lies in the range of 8–15, providing an effective balance among representational capacity, computational efficiency, and detection accuracy. These results demonstrate that the proposed framework is not sensitive to a specific value of K, and a moderate number of etalons is sufficient to achieve a favorable tradeoff between representation capability and computational efficiency. To jointly consider computational cost and performance, all experiments involving HACL-DC in this article adopt $K = 10$ when the number of training samples is 150 or 200, while for smaller training sets of 50–100 samples, $K = 5$ is adopted. These findings further confirm the robustness and practical applicability of the SWCA module within the HACL-DC framework.

V. Conclusion

In this article, we proposed HACL-DC, a novel open-set HSIC framework that integrates hardness-aware contrastive learning with soft-label-weighted data condensation. The proposed framework was designed to address several critical challenges in existing open-set hyperspectral classification methods, including insufficient exploitation of hard samples, weak intraclass compactness, and the inability of conventional prototype-based approaches to effectively model complex multimodal feature distributions. Specifically, the proposed HSCWM module improves feature discriminability by assigning adaptive weights to hard positive and hard negative sample pairs, thereby simultaneously enhancing intraclass compactness and interclass separability in the hyperspherical feature space. Furthermore, the proposed SWCA module constructs structurally representative etalons through soft-label assignment, dual-parameter optimization, and adaptive etalon refinement, enabling more robust characterization of complex class distributions and more reliable distance-based OOD detection.

Extensive experiments conducted on the Indian Pines, WHU-Hi-LongKou, and WHU-Hi-HanChuan datasets demonstrate that the proposed HACL-DC consistently achieves superior performance compared with representative postprocessing and training-based open-set methods. In particular, the proposed method achieves substantial improvements in unknown-class detection accuracy while maintaining strong closed-set classification performance across different training sample settings. These results indicate that jointly optimizing discriminative feature learning and structurally aware class representation is highly beneficial for open-set HSIC.

Nevertheless, several aspects can be further improved in future work. For example, although the sensitivity analysis demonstrates the robustness of the proposed framework, developing more adaptive hyperparameter selection strategies remains a promising direction. Moreover, although the threshold sensitivity analysis demonstrates the robustness of HACL-DC under different threshold settings, the current threshold calibration strategy still relies on a predefined TPR constraint, which may not be optimal for all application scenarios. Developing more rigorous and adaptive threshold calibration strategies remains an important direction for future research. In addition, incorporating semantic prior information or cross-scene adaptation strategies may further improve the generalization capability of the proposed framework in more complex real-world scenarios. Future work will also investigate more efficient adaptive condensation strategies, automatic etalon selection mechanisms, and semantic-guided representation learning to further enhance the robustness and scalability of open-set HSIC.

References

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] D. Wu and D.-W. Sun, "Advanced applications of hyperspectral imaging technology for food quality and safety analysis and assessment: A review—Part I: Fundamentals," *Innov. Food Sci. Emerg. Technol.*, vol. 19, pp. 1–14, Jul. 2013.

[3] B. Fu et al., "Quantifying vegetation species functional traits along hydrologic gradients in Karst wetland based on 3D mapping with UAV hyperspectral point cloud," *Remote Sens. Environ.*, vol. 307, Jun. 2024, Art. no. 114160.

[4] R. Thoreau, V. Achard, L. Risser, B. Berthelot, and X. Briottet, "Active learning for hyperspectral image classification: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 256–278, Sep. 2022.

[5] Y. Du, X. Li, L. Shi, F. Li, and T. Xu, "A prototype network for hyperspectral image open-set classification based on feature invariance and weighted Pearson distance measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506917.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.

[6] J. M. Amigo, H. Babamoradi, and S. Elcoroaristizabal, "Hyperspectral image analysis. A tutorial," *Analytica Chim. Acta*, vol. 896, pp. 34–51, Oct. 2015.
[7] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.
[8] C. Cariou and K. Chehdi, "A new k-nearest neighbor density-based clustering method and its application to hyperspectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 6161–6164.
[9] P. Burai, B. Deák, O. Valkó, and T. Tomor, "Classification of herbaceous vegetation using airborne hyperspectral imagery," *Remote Sens.*, vol. 7, no. 2, pp. 2046–2066, Feb. 2015.
[10] B. Asghari Beirami and M. Mokhtarzade, "Band grouping SuperPCA for feature extraction and extended morphological profile production from hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 11, pp. 1953–1957, Nov. 2020.
[11] S. Jia et al., "Flexible Gabor-based superpixel-level unsupervised LDA for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10394–10409, Dec. 2021.
[12] L. Fang, S. Li, X. Kang, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification via multiscale adaptive sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7738–7749, Dec. 2014.
[13] Z. Xue, P. Du, J. Li, and H. Su, "Simultaneous sparse graph embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6114–6133, Nov. 2015.
[14] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021.
[15] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
[16] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.
[17] G. Tejasree and L. Agilandeeswari, "Land use/land cover (LULC) classification using deep-LSTM for hyperspectral images," *Egyptian J. Remote Sens. Space Sci.*, vol. 27, no. 1, pp. 52–68, Mar. 2024.
[18] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.
[19] G. Wu, M. A. A. Al-Qaness, D. Al-Alimi, A. Dahou, M. Abd Elaziz, and A. A. Ewees, "Hyperspectral image classification using graph convolutional network: A comprehensive review," *Expert Syst. Appl.*, vol. 257, Dec. 2024, Art. no. 125106.
[20] H. Qi et al., "SAM-GAN: An improved DCGAN for Rice seed viability determination using near-infrared hyperspectral imaging," *Comput. Electron. Agricult.*, vol. 216, Jan. 2024, Art. no. 108473.
[21] V. Kumar, R. S. Singh, M. Rambabu, and Y. Dua, "Deep learning for hyperspectral image classification: A survey," *Comput. Sci. Rev.*, vol. 53, Aug. 2024, Art. no. 100658.
[22] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.
[23] B. Zhang, Y. Chen, S. Xiong, and X. Lu, "Hyperspectral image classification via cascaded spatial cross-attention network," *IEEE Trans. Image Process.*, vol. 34, pp. 899–913, 2025.
[24] Y. Liu, J. Hou, Y. Peng, Y. Xu, and T. Jiang, "Hyperspectral open set classification towards deep networks based on boxplot," in *Proc. IOP Conf., Earth Environ. Sci.*, Mar. 2021, vol. 693, no. 1, Art. no. 012085.
[25] B. Alwadei, M. L. Mekhalfi, Y. Bazi, M. M. Al Rahhal, and M. Zuair, "Open set classification of hyperspectral images with energy models," *Int. J. Remote Sens.*, vol. 44, no. 24, pp. 7876–7888, Dec. 2023.
[26] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.
[27] L. Fang, D. Zhu, J. Yue, B. Zhang, and M. He, "Geometric-spectral reconstruction learning for multi-source open-set classification with hyperspectral and LiDAR data," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 10, pp. 1892–1895, Oct. 2022.
[28] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.
[29] F. Ji et al., "Spectral–spatial evidential learning network for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503617.
[30] L. Ma, Y. Yang, G. Wang, Z. Li, W. Sun, and Q. Du, "Open-set domain adaptation for hyperspectral image classification based on reconstruction discrepancy and feature alignment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 24829–24847, 2025.
[31] B. Xi, W. Zhang, J. Li, R. Song, and Y. Li, "HyperCASR: Spectral–spatial open-set recognition with category-aware semantic reconstruction for hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 34, pp. 7642–7655, 2025.
[32] Z. Xie, P. Duan, W. Liu, X. Kang, X. Wei, and S. Li, "Feature consistency-based prototype network for open-set hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9286–9296, Jul. 2024.
[33] H. Xu, W. Chen, C. Tan, H. Ning, H. Sun, and W. Xie, "Orientational clustering learning for open-set hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[34] M. Chen et al., "Fractional Fourier-based frequency-spatial–spectral prototype network for agricultural hyperspectral image open-set classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514014.
[35] S. Feng, S. Wang, C. Xu, C. Zhao, W. Li, and R. Tao, "Fractional-domain information-enhanced hyperspherical prototype learning method for hyperspectral image open-set classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515117.
[36] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and J. Zhou, "Language-enhanced dual-level contrastive learning network for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508114.
[37] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521213.
[38] D. J. Trosten, R. Chakraborty, S. Løkse, K. K. Wickstrøm, R. Jenssen, and M. C. Kampffmeyer, "Hubs and hyperspheres: Reducing hubness and improving transductive few-shot learning with hyperspherical embeddings," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7527–7536.
[39] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[40] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Scalable recurrent neural network for hyperspectral image classification," *J. Supercomput.*, vol. 76, no. 11, pp. 8866–8882, Nov. 2020.
[41] H. Lu et al., "Learning with mixture of prototypes for out-of-distribution detection," in *Proc. Int. Conf. Learn. Represent.*, 2024, pp. 1–23.
[42] Y. Sun, C. Guo, and Y. Li, "Out-of-distribution detection with rectified activations," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 144–157.
[43] A. Djurisic, N. Bozanic, A. Ashok, and R. Liu, "Extremely simple activation shaping for out-of-distribution detection," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–22.
[44] J. Zhang et al., "Out-of-distribution detection based on in-distribution data patterns memorization with modern Hopfield energy," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–19.
[45] L. Liu and Y. Qin, "Detecting out-of-distribution through the lens of neural collapse," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 15424–15433.
[46] K. Xu, R. Chen, G. Franchi, and A. Yao, "Scaling for training time and post-hoc out-of-distribution detection enhancement," in *Proc. Int. Conf. Learn. Represent.*, 2023, pp. 1–14.
[47] S. Regmi, "Going beyond conventional OOD detection," 2025, *arXiv:2411.10794*.



Hanyue Wang received the B.S. degree in intelligence science and technology from North China University of Science and Technology, Tangshan, Hebei, China, in 2024. She is currently pursuing the master's degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

Her research interests include hyperspectral image processing and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.



Wei Liu received the B.S. degree in computer science and technology from Xidian University, Xi'an, Shaanxi, China, in 2024, where he is currently pursuing the master's degree with the School of Artificial Intelligence.

His research interests include hyperspectral image processing and out-of-distribution detection.



Jing Zhao received the bachelor's and master's degrees in information and communication engineering from Xidian University, Xi'an, China, in 2003 and 2006, respectively, where she is currently pursuing the Ph.D. degree in computer science and technology.

Her research interests include hyperspectral images classification, machine learning, and deep learning.



Xianghai Cao (Member, IEEE) received the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2008.

Since 2008, he has been with Xidian University, where he is currently a Professor with the School of Artificial Intelligence. His research interests include hyperspectral image processing, pattern recognition, and machine learning.



Licheng Jiao (Life Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Artificial Intelligence, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China. He is in charge of about 40 important scientific research projects and has published more than 20 monographs and 100 articles in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a fellow of the IEEE Xi'an Section Execution Committee, the Chairperson of the Awards and Recognition Committee, the Vice Board Chairperson of Chinese Association of Artificial Intelligence, a Councilor of Chinese Institute of Electronics, a Committee Member of Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:57:58 UTC from IEEE Xplore. Restrictions apply.